

## Assignment 1 : Combinational Logic Design

The GitHub Link contains all the source codes and VCD dumps of the files and stuff :

1. [https://github.com/ssbagi/IBM\\_OpenPOWER\\_and\\_SoC\\_Design/tree/main/DAY2/Assignment201](https://github.com/ssbagi/IBM_OpenPOWER_and_SoC_Design/tree/main/DAY2/Assignment201)
2. README file on this Assignment :

### Problems :

1. Design and simulate all the six fundamental logic gates OR, NOT, NAND, NOR, XOR, XNOR in Vivado/ Modelsim in dataflow and structural modeling styles.

### Dataflow Modelling Code :

```
/*
```

```
Dataflow modelling of the Following gates
```

```
1. AND
```

```
2. NAND
```

```
3. OR
```

```
4. NOR
```

```
5. NOT
```

```
6. XOR
```

```
7. XNOR
```

```
*/
```

```
module and_gate(a, b, y);
```

```
    input a, b;
```

```
    output y;
```

```
    assign y = a & b;
```

```
    initial $monitor("Time = %0t | AND GATE y = a & b | y = %b", $time, y);
```

```
endmodule
```

```
module nand_gate(a, b, y);  
    input a, b;  
    output y;  
    assign y = ~(a & b);  
    initial $monitor("Time = %0t | NAND GATE y = ~(a & b) | y = %b", $time, y);  
endmodule
```

```
module or_gate(a, b, y);  
    input a, b;  
    output y;  
    assign y = a | b;  
    initial $monitor("Time = %0t | OR GATE y = (a | b) | y = %b", $time, y);  
endmodule
```

```
module nor_gate(a, b, y);  
    input a, b;  
    output y;  
    assign y = ~(a | b);  
    initial $monitor("Time = %0t | NOR GATE y = ~(a | b) | y = %b", $time, y);  
endmodule
```

```
module not_gate(a, y);  
    input a;  
    output y;  
    assign y = ~a;  
    initial $monitor("Time = %0t | NOT GATE y = (~a) | y = %b", $time, y);  
endmodule
```

```

module xor_gate(a, b, y);
    input a, b;
    output y;
    assign y = a ^ b;
    initial $monitor("Time = %0t | XOR GATE y = (a ^ b) | y = %b", $time, y);
endmodule

```

```

module xnor_gate(a, b, y);
    input a, b;
    output y;
    assign y = ~(a ^ b);
    initial $monitor("Time = %0t | XNOR GATE y = ~(a ^ b) | y = %b", $time, y);
endmodule

```

### **Structural Modelling Code :**

```

module and_gate(a, b, y);
    input a, b;
    output y;
    and a1(y, a, b);
    initial $monitor("Time = %0t | AND GATE y = a & b | y = %b", $time, y);
endmodule

```

```

module nand_gate(a, b, y);
    input a, b;
    output y;
    nand n1(y, a, b);

```

```
    initial $monitor("Time = %0t | NAND GATE  $y = \sim(a \& b)$  |  $y = \%b$ ", $time, y);  
endmodule
```

```
module or_gate(a, b, y);  
    input a, b;  
    output y;  
    or o1(y, a, b);  
    initial $monitor("Time = %0t | OR GATE  $y = (a \mid b)$  |  $y = \%b$ ", $time, y);  
endmodule
```

```
module nor_gate(a, b, y);  
    input a, b;  
    output y;  
    nor n2(y, a, b);  
    initial $monitor("Time = %0t | NOR GATE  $y = \sim(a \mid b)$  |  $y = \%b$ ", $time, y);  
endmodule
```

```
module not_gate(a, y);  
    input a;  
    output y;  
    not n3(y, a);  
    initial $monitor("Time = %0t | NOT GATE  $y = (\sim a)$  |  $y = \%b$ ", $time, y);  
endmodule
```

```
module xor_gate(a, b, y);  
    input a, b;  
    output y;  
    xor x1(y, a, b);
```

```

        initial $monitor("Time = %0t | XOR GATE y = (a ^ b) | y = %b", $time, y);
    endmodule

module xnor_gate(a, b, y);
    input a, b;
    output y;
    xnor x2(y, a, b);
    initial $monitor("Time = %0t | XNOR GATE y = ~(a ^ b) | y = %b", $time, y);
endmodule

```

Testbench :

```

`ifndef DATAFLOW
    `include "logic_gates_dataflow.v"
`else
    `include "logic_gates_structure.v"
`endif

module logic_gates_tb;
    reg a, b;
    wire y_and, y_nand, y_or, y_nor, y_not, y_xor, y_xnor;

    and_gate u_and (a, b, y_and);
    nand_gate u_nand (a, b, y_nand);
    or_gate u_or (a, b, y_or);
    nor_gate u_nor (a, b, y_nor);
    not_gate u_not (a, y_not);
    xor_gate u_xor (a, b, y_xor);

```

```
xnor_gate u_xnor (a, b, y_xnor);
```

```
initial begin
```

```
    $dumpfile("logic_gates_tb.vcd");
```

```
    $dumpvars(0, logic_gates_tb);
```

```
    a = 0; b = 0;
```

```
    #10 a = 0; b = 1;
```

```
    #10 a = 1; b = 0;
```

```
    #10 a = 1; b = 1;
```

```
    #10 $finish;
```

```
end
```

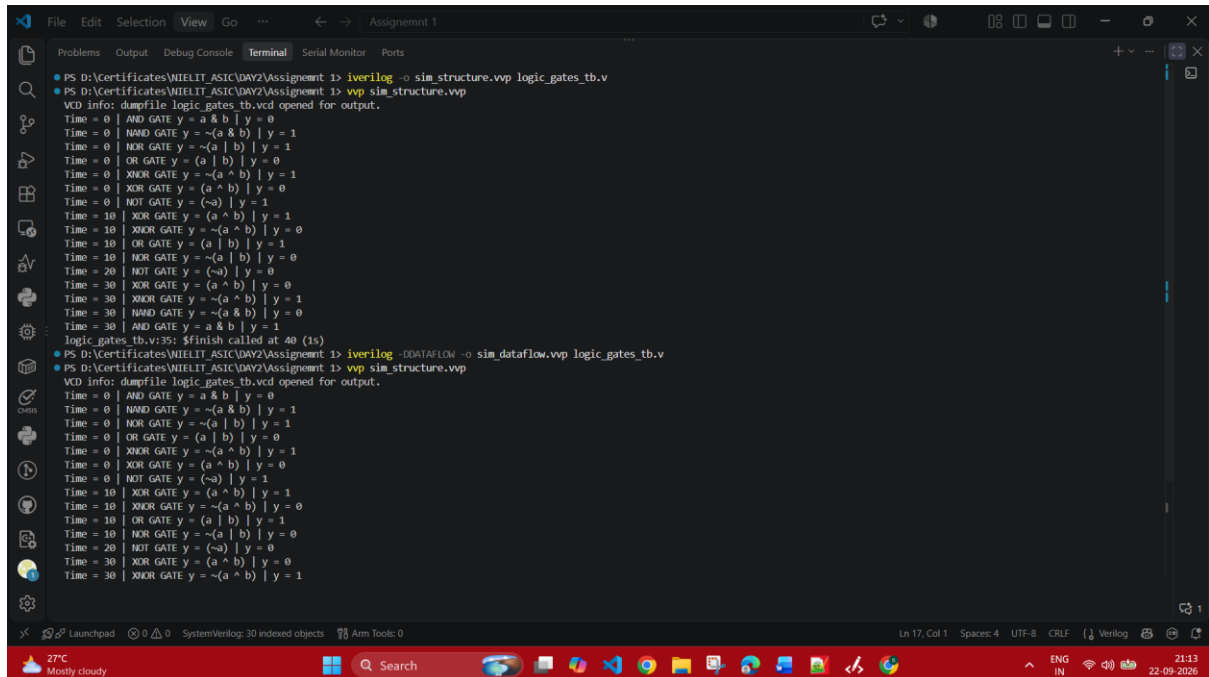
```
endmodule
```

## Compilation :

iverilog -o sim\_structure.vvp logic\_gates\_tb.v # structural (default)

iverilog -DDATAFLOW -o sim\_dataflow.vvp logic\_gates\_tb.v # dataflow

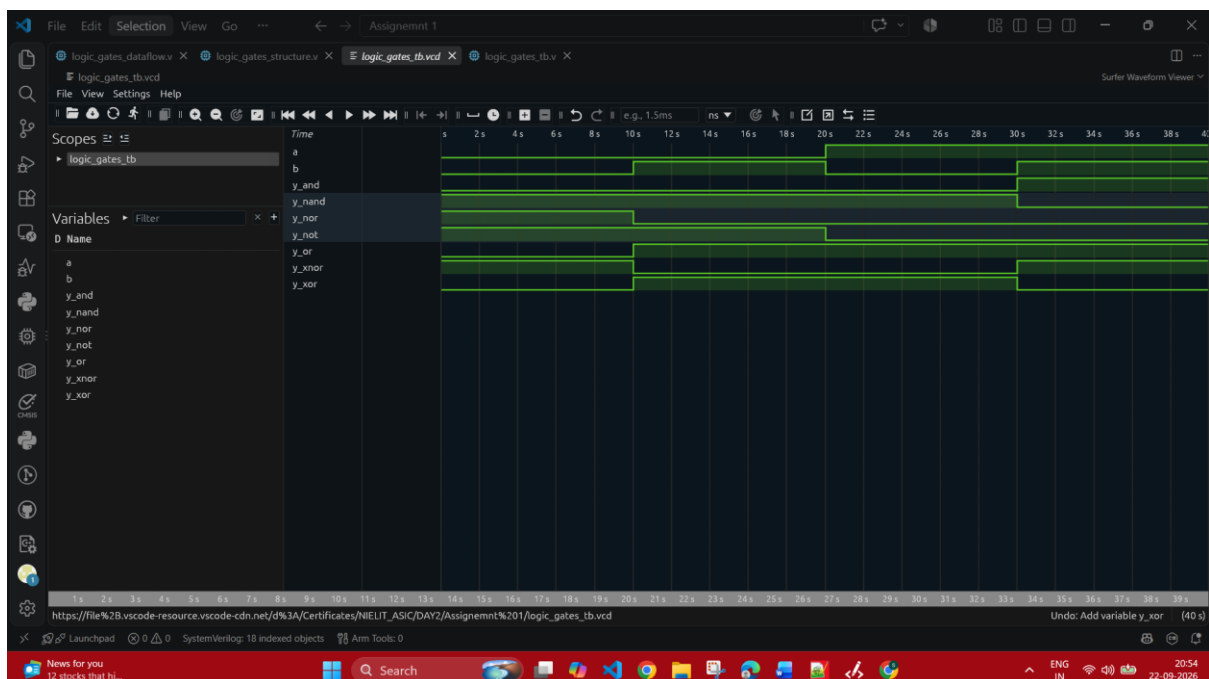
vvp sim\_structure.vvp



```
PS D:\Certificates\NIELIT_ASIC\DAY2\Assignemnt 1> iverilog -o sim_structure.vvp logic_gates_tb.v
PS D:\Certificates\NIELIT_ASIC\DAY2\Assignemnt 1> vvp sim_structure.vvp
VCD info: dumpfile logic_gates_tb.vcd opened for output.
Time = 0 | AND GATE y = a & b | y = 0
Time = 0 | NAND GATE y = ~(a & b) | y = 1
Time = 0 | NOR GATE y = ~(a | b) | y = 1
Time = 0 | OR GATE y = (a | b) | y = 0
Time = 0 | XOR GATE y = ~(a ^ b) | y = 1
Time = 0 | XOR GATE y = (a ^ b) | y = 0
Time = 0 | NOT GATE y = (~a) | y = 1
Time = 10 | XOR GATE y = (a ^ b) | y = 1
Time = 10 | XOR GATE y = ~(a ^ b) | y = 0
Time = 10 | OR GATE y = (a | b) | y = 1
Time = 10 | NOR GATE y = ~(a | b) | y = 0
Time = 20 | NOT GATE y = (~a) | y = 0
Time = 30 | XOR GATE y = (a ^ b) | y = 0
Time = 30 | XOR GATE y = ~(a ^ b) | y = 1
Time = 30 | NAND GATE y = ~(a & b) | y = 0
Time = 30 | AND GATE y = a & b | y = 1
logic_gates_tb.v:35: $finish called at 40 (1s)
PS D:\Certificates\NIELIT_ASIC\DAY2\Assignemnt 1> iverilog -DDATAFLOW -o sim_dataflow.vvp logic_gates_tb.v
PS D:\Certificates\NIELIT_ASIC\DAY2\Assignemnt 1> vvp sim_dataflow.vvp
VCD info: dumpfile logic_gates_tb.vcd opened for output.
Time = 0 | AND GATE y = a & b | y = 0
Time = 0 | NAND GATE y = ~(a & b) | y = 1
Time = 0 | NOR GATE y = ~(a | b) | y = 1
Time = 0 | OR GATE y = (a | b) | y = 0
Time = 0 | XOR GATE y = ~(a ^ b) | y = 1
Time = 0 | XOR GATE y = (a ^ b) | y = 0
Time = 0 | NOT GATE y = (~a) | y = 1
Time = 10 | XOR GATE y = (a ^ b) | y = 1
Time = 10 | XOR GATE y = ~(a ^ b) | y = 0
Time = 10 | OR GATE y = (a | b) | y = 1
Time = 10 | NOR GATE y = ~(a | b) | y = 0
Time = 20 | NOT GATE y = (~a) | y = 0
Time = 30 | XOR GATE y = (a ^ b) | y = 0
Time = 30 | XOR GATE y = ~(a ^ b) | y = 1
```

## From the Testbench driven :

Every 10ns there is event change in the inputs a and b. The output changes based on the gates and stuff.



Q2. Design a 2:1 Multiplexer using Verilog HDL in different modeling styles.

- a. Dataflow Modeling using continuous assignment (assign) statement.

```
module mux_2_to_1_behavioral (  
    input wire a,  
    input wire b,  
    input wire sel,  
    output reg y  
);
```

```
    always @(sel, a, b) begin  
        if (sel)  
            y = a;  
        else  
            y = b;  
    end
```

```
endmodule
```

- b. Dataflow Modeling using the conditional (?:) operator.

```
module mux_2_to_1_behavioral (  
    input wire a,  
    input wire b,  
    input wire sel,  
    output reg y  
);
```

```
    always @(sel, a, b) begin  
        if (sel)  
            y = a;  
        else  
            y = b;  
    end
```

```
endmodule
```

- c. Behavioural Modeling using always block.

```
module mux_2_to_1_behavioral_always (  
    input wire a,  
    input wire b,  
    input wire sel,
```

```

        output reg y
    );

    always @(a, b, sel) begin
        if (sel)
            y = a;
        else
            y = b;
        end

endmodule

```

- d. Behavioural Modeling using if-else statements.

```

module mux_2_to_1_behavioral_ifelse (
    input wire a,
    input wire b,
    input wire sel,
    output reg y
);

    always @(a, b, sel) begin
        if (sel)
            y = a;
        else
            y = b;
        end

endmodule

```

- e. Behavioural Modeling using case statements.

```

module mux_2_to_1_behavioral_case (
    input wire a,
    input wire b,
    input wire sel,
    output reg y
);

    always @(*) begin
        case (sel)
            1'b0 : y = a;

```

```

        1'b1 : y = b;
    endcase
end

endmodule

```

- f. Gate-Level Modeling using logic primitives (and, or, not).

```

module mux_2_to_1_gate_level (
    input wire a,
    input wire b,
    input wire sel,
    output wire y
);

    wire sel_n, y1, y2;

    not(sel_n, sel);
    and(y1, sel_n, a);
    and(y2, sel, b);
    or(y, y1, y2);

endmodule

```

Compilation of the Mux related Verilog Files :

```
File Edit Selection View Go ... Assignmmt 1
Problems Output Debug Console Terminal Serial Monitor Ports
PS D:\Certificates\WIELIT ASIC\DAY2\Assignmmt 1> iverilog -o sim_structure.vvp logic_gates.tb.v; vvp sim_structure.vvp
PS D:\Certificates\WIELIT ASIC\DAY2\Assignmmt 1> iverilog -o sim_mux_dataflow.vvp mux_2_to_1_dataflow.tb.v; vvp sim_mux_dataflow.vvp
VCD info: dumpfile mux_2_to_1_dataflow_tb.vcd opened for output.
Time = 0 | a=0 b=0 sel=0 | y=0
Time = 10 | a=0 b=1 sel=0 | y=0
Time = 20 | a=0 b=1 sel=1 | y=1
Time = 30 | a=1 b=0 sel=0 | y=1
Time = 40 | a=1 b=0 sel=1 | y=0
mux_2_to_1_dataflow.tb.v:19: $finish called at 50 (1s)
PS D:\Certificates\WIELIT ASIC\DAY2\Assignmmt 1> iverilog -o sim_mux_dataflow_ternary.vvp mux_2_to_1_dataflow_ternary.tb.v; vvp sim_mux_dataflow_ternary.vvp
VCD info: dumpfile mux_2_to_1_dataflow_ternary_tb.vcd opened for output.
Time = 0 | a=0 b=0 sel=0 | y=0
Time = 10 | a=0 b=1 sel=0 | y=1
Time = 20 | a=0 b=1 sel=1 | y=0
Time = 30 | a=1 b=0 sel=0 | y=0
Time = 40 | a=1 b=0 sel=1 | y=1
mux_2_to_1_dataflow_ternary.tb.v:19: $finish called at 50 (1s)
PS D:\Certificates\WIELIT ASIC\DAY2\Assignmmt 1> iverilog -o sim_mux_behavioral.vvp mux_2_to_1_behavioral.tb.v; vvp sim_mux_behavioral.vvp
VCD info: dumpfile mux_2_to_1_behavioral_tb.vcd opened for output.
Time = 0 | a=0 b=0 sel=0 | y=0
Time = 10 | a=0 b=1 sel=0 | y=1
Time = 20 | a=0 b=1 sel=1 | y=0
Time = 30 | a=1 b=0 sel=0 | y=0
Time = 40 | a=1 b=0 sel=1 | y=1
mux_2_to_1_behavioral.tb.v:19: $finish called at 50 (1s)
PS D:\Certificates\WIELIT ASIC\DAY2\Assignmmt 1> iverilog -o sim_mux_behavioral_ifelse.vvp mux_2_to_1_behavioral_ifelse.tb.v; vvp sim_mux_behavioral_ifelse.vvp
VCD info: dumpfile mux_2_to_1_behavioral_ifelse_tb.vcd opened for output.
Time = 0 | a=0 b=0 sel=0 | y=0
Time = 10 | a=0 b=1 sel=0 | y=1
Time = 20 | a=0 b=1 sel=1 | y=0
Time = 30 | a=1 b=0 sel=0 | y=0
Time = 40 | a=1 b=0 sel=1 | y=1
mux_2_to_1_behavioral_ifelse.tb.v:19: $finish called at 50 (1s)
PS D:\Certificates\WIELIT ASIC\DAY2\Assignmmt 1> iverilog -o sim_mux_behavioral_case.vvp mux_2_to_1_behavioral_case.tb.v; vvp sim_mux_behavioral_case.vvp
VCD info: dumpfile mux_2_to_1_behavioral_case_tb.vcd opened for output.
Time = 0 | a=0 b=0 sel=0 | y=0
Time = 10 | a=0 b=1 sel=0 | y=1
Time = 20 | a=0 b=1 sel=1 | y=1
PS D:\Certificates\WIELIT ASIC\DAY2\Assignmmt 1>
SystemVerilog: 30 indexed objects Arm Tools: 0
Ln 24, Col 1 Spaces: 4 UTF-8 CRLF Verilog
27°C Mostly cloudy 21:17 22-09-2026
```

```
File Edit Selection View Go ... Assignmmt 1
Problems Output Debug Console Terminal Serial Monitor Ports
PS D:\Certificates\WIELIT ASIC\DAY2\Assignmmt 1> iverilog -o sim_mux_behavioral.vvp mux_2_to_1_behavioral.tb.v; vvp sim_mux_behavioral.vvp
VCD info: dumpfile mux_2_to_1_behavioral_tb.vcd opened for output.
Time = 0 | a=0 b=0 sel=0 | y=0
Time = 10 | a=0 b=1 sel=0 | y=1
Time = 20 | a=0 b=1 sel=1 | y=0
Time = 30 | a=1 b=0 sel=0 | y=0
Time = 40 | a=1 b=0 sel=1 | y=1
mux_2_to_1_behavioral.tb.v:19: $finish called at 50 (1s)
PS D:\Certificates\WIELIT ASIC\DAY2\Assignmmt 1> iverilog -o sim_mux_behavioral_ifelse.vvp mux_2_to_1_behavioral_ifelse.tb.v; vvp sim_mux_behavioral_ifelse.vvp
VCD info: dumpfile mux_2_to_1_behavioral_ifelse_tb.vcd opened for output.
Time = 0 | a=0 b=0 sel=0 | y=0
Time = 10 | a=0 b=1 sel=0 | y=1
Time = 20 | a=0 b=1 sel=1 | y=0
Time = 30 | a=1 b=0 sel=0 | y=0
Time = 40 | a=1 b=0 sel=1 | y=1
mux_2_to_1_behavioral_ifelse.tb.v:19: $finish called at 50 (1s)
PS D:\Certificates\WIELIT ASIC\DAY2\Assignmmt 1> iverilog -o sim_mux_behavioral_case.vvp mux_2_to_1_behavioral_case.tb.v; vvp sim_mux_behavioral_case.vvp
VCD info: dumpfile mux_2_to_1_behavioral_case_tb.vcd opened for output.
Time = 0 | a=0 b=0 sel=0 | y=0
Time = 10 | a=0 b=1 sel=0 | y=1
Time = 20 | a=0 b=1 sel=1 | y=1
Time = 30 | a=1 b=0 sel=0 | y=1
Time = 40 | a=1 b=0 sel=1 | y=0
mux_2_to_1_behavioral_case.tb.v:19: $finish called at 50 (1s)
PS D:\Certificates\WIELIT ASIC\DAY2\Assignmmt 1> iverilog -o sim_mux_gate_level.vvp mux_2_to_1_gate_level.tb.v; vvp sim_mux_gate_level.vvp
VCD info: dumpfile mux_2_to_1_gate_level_tb.vcd opened for output.
Time = 0 | a=0 b=0 sel=0 | y=0
Time = 10 | a=0 b=1 sel=0 | y=0
Time = 20 | a=0 b=1 sel=1 | y=1
Time = 30 | a=1 b=0 sel=0 | y=1
Time = 40 | a=1 b=0 sel=1 | y=0
mux_2_to_1_gate_level.tb.v:19: $finish called at 50 (1s)
PS D:\Certificates\WIELIT ASIC\DAY2\Assignmmt 1>
SystemVerilog: 30 indexed objects Arm Tools: 0
Ln 24, Col 1 Spaces: 4 UTF-8 CRLF Verilog
27°C Mostly cloudy 21:18 22-09-2026
```

The VCD dumps of the Mux related ones. All the VCD dump are same.

